

The Problem

The `@capgo/capacitor-health` plugin provides a unified API for reading health data on both iOS (HealthKit) and Android (Health Connect). However, the plugin has a significant issue on Android: it maps the `calories` data type to `ActiveCaloriesBurnedRecord` instead of `TotalCaloriesBurnedRecord`.

In Health Connect, these are two separate record types:

- `TotalCaloriesBurnedRecord` — All calories burned (what most fitness trackers write)
- `ActiveCaloriesBurnedRecord` — Only calories from deliberate activity (many apps, including Samsung Health, don't write this separately)

So when you ask the plugin for "calories," you get active calories (often zero from Samsung Health) instead of total calories. The plugin also has no support at all for two other Health Connect record types:

- `ExerciseSessionRecord` — Workout sessions with start/end times, duration, and exercise type
- `BasalMetabolicRateRecord` — Resting metabolic rate

What the Patch File Does

The patch script (`scripts/patch-health-plugin.cjs`) runs after `npm install` and modifies two Kotlin source files inside the plugin's `node_modules` directory:

1. `HealthDataType.kt` — Rewrites the enum to add `ACTIVE_CALORIES`, `EXERCISE`, and `BMR` as new data types, and correctly maps `CALORIES` to `TotalCaloriesBurnedRecord` instead of `ActiveCaloriesBurnedRecord`.
2. `HealthManager.kt` — Adds the necessary imports and read/write blocks so the plugin knows how to query `ExerciseSessionRecord` and `BasalMetabolicRateRecord` from Health Connect, and fixes the calorie type mapping in the read/write logic.

After patching, the plugin supports these data types on Android:

- `steps` → `StepsRecord`
- `calories` → `TotalCaloriesBurnedRecord` (fixed from `ActiveCaloriesBurnedRecord`)
- `activeCalories` → `ActiveCaloriesBurnedRecord` (new, separate type)
- `exercise` → `ExerciseSessionRecord` (new)
- `bmr` → `BasalMetabolicRateRecord` (new)
- `heartRate` → `HeartRateRecord`
- `weight` → `WeightRecord`
- `distance` → `DistanceRecord`

Prompt for Another Replit User

Here's a prompt your friend can use:

Prompt to give to a Replit Agent:

I am building a health data app using `@capgo/capacitor-health` for Android (Health Connect). The plugin has a bug: it maps the "calories" data type to `ActiveCaloriesBurnedRecord` instead of `TotalCaloriesBurnedRecord`. Samsung Health and most fitness apps write total calories, not active calories separately, so the plugin returns zero or incorrect calorie data.

I also need support for `ExerciseSessionRecord` (workout sessions with duration) and `BasalMetabolicRateRecord` (resting metabolic rate), which the plugin does not support at all.

Please create a post-install patch script at `scripts/patch-health-plugin.cjs` that modifies the plugin's Android Kotlin source files after npm install. The script should:

1. Rewrite `HealthDataType.kt` (located in `node_modules/@capgo/capacitor-health/android/src/main/java/app/capgo/plugin/health/`) to:
 - Map `CALORIES` to `TotalCaloriesBurnedRecord` instead of `ActiveCaloriesBurnedRecord`
 - Add a new `ACTIVE_CALORIES` type mapped to `ActiveCaloriesBurnedRecord`
 - Add a new `EXERCISE` type mapped to `ExerciseSessionRecord` with unit "minute"
 - Add a new `BMR` type mapped to `BasalMetabolicRateRecord` with unit "kilocalorie"
2. Patch `HealthManager.kt` to:
 - Add imports for `TotalCaloriesBurnedRecord`, `ExerciseSessionRecord`, and `BasalMetabolicRateRecord`
 - Fix the `CALORIES` read/write blocks to use `TotalCaloriesBurnedRecord`
 - Add read/write blocks for `ACTIVE_CALORIES`, `EXERCISE`, and `BMR`
 - For exercise, read the duration in minutes calculated from start and end times
 - For BMR, handle it as an instantaneous record (uses `record.time` instead of `record.startTime/record.endTime`)
3. Add a `"postinstall": "node scripts/patch-health-plugin.cjs"` script to `package.json` so the patch runs automatically after every `npm install`.

Also update the Android manifest to include these permissions: `READ_TOTAL_CALORIES_BURNED`, `READ_ACTIVE_CALORIES_BURNED`, `READ_EXERCISE`, and `READ_BASAL_METABOLIC_RATE`.